

WEEK 2 – FINAL PROJECT REPORT

Footprinting with Kali Linux Tools & Zenmap-Based Network Scanning

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Program	Cybersecurity Internship / Workflow Program
Week	Week 2
Elective	W2-PM1 – Footprinting with Multiple Kali Tools
Essential	W2-PM5 – Zenmap-Based Network Scanning
Final Module	W2-PM-FINAL – Detailed Report

This report documents the practical work completed for the selected Week 2 modules. Testing was performed for learning and authorized training purposes.

Prepared as a practical training report

1. Executive Summary

The Week 2 project focused on two areas of introductory cybersecurity reconnaissance. The first module, W2-PM1, involved using multiple Kali Linux command-line tools to collect publicly observable information about a target domain. The second module, W2-PM5, used Zenmap/Nmap to perform basic host discovery on a private lab network.

The footprinting stage used six tools/commands: WHOIS, WhatWeb, WAFW00F, cURL, nslookup, and dnsrecon. The collected results provided information about domain registration, web technologies, HTTP behavior, DNS resolution, and web-application protection indicators.

For the network-scanning stage, Zenmap was configured with the Ping Scan profile against 192.168.56.1. The scan identified the host as reachable/alive and displayed it in the topology view alongside the local host.

2. Objectives

- Practice passive and lightweight technical footprinting using Kali Linux tools.
- Identify publicly observable domain, DNS, HTTP, and web-technology information.
- Understand the purpose and output of common reconnaissance tools.
- Perform basic host discovery using Zenmap/Nmap in a controlled lab environment.
- Document practical results clearly and professionally.

3. Scope and Methodology

The footprinting exercises were performed against the training target **networkwalks.com**, while the Zenmap exercise used the private lab address **192.168.56.1**. The activities were limited to reconnaissance and host discovery commands demonstrated as part of the internship module.

No exploitation, credential attacks, persistence, or destructive activity was performed. The purpose of the exercises was to understand how security professionals collect and interpret reconnaissance data.

4. W2-PM1 – Footprinting with Multiple Kali Tools

Six Kali Linux tools/commands were used to gather different types of information. Each tool contributes a different perspective, so the outputs are more useful when interpreted together.

Tool / Command	Purpose	Observed Result / Key Information
WHOIS	Domain registration information	Returned registrar, domain status, name servers, creation/update/expiry information and registrar details
WhatWeb	Web technology fingerprinting	Identified technologies and headers including Apache, WordPress, jQuery and other web components
WAFW00F	WAF/security-layer detection	Reported that the site appeared to be behind a WAF or another security solution and observed characteristics
cURL (-I)	HTTP header inspection	Returned HTTP response headers including a 301 redirect, Apache server information and the media type
nslookup	DNS resolution	Resolved networkwalks.com to the IPv4 address 192.232.216.135 using the configured DNS resolver
dnsrecon	DNS enumeration	Completed domain enumeration; output included DNSSEC/SOA timeout or resolution messages

5. Detailed W2-PM1 Observations

5.1 WHOIS

WHOIS was used to examine registration-related information for the training domain. The captured output showed the domain name, registrar information, registration dates, domain status values, and name servers. This demonstrates how WHOIS can provide useful context during the initial reconnaissance phase.

```
Command: whois networkwalks.com
```

5.2 WhatWeb

WhatWeb was used for web-technology fingerprinting. The observed output identified Apache as the web server and showed technologies/components including WordPress and jQuery. It also exposed several HTTP headers and website metadata. Such information helps a security analyst understand the technologies present before further authorized testing.

```
Command: whatweb networkwalks.com
```

5.3 WAFW00F

WAFW00F was used to check whether the website appeared to be protected by a web application firewall or similar filtering layer. The captured result indicated that the site appeared to be behind a WAF or another security solution, with connection-level blocking observed. This is an important reconnaissance observation because defensive controls can influence how web requests are handled.

```
Command: wafw00f networkwalks.com
```

5.4 cURL

cURL was used to inspect HTTP response headers. The captured response included an HTTP 301 redirect and Apache server information, with the response pointing toward the HTTPS version of the website. Header inspection is a simple way to understand how a web server responds without loading the full page.

```
Command: curl -I networkwalks.com
```

5.5 nslookup

nslookup successfully resolved networkwalks.com and returned the IPv4 address **192.232.216.135**. This demonstrates basic DNS resolution and provides an IP address that can be used as a reference for authorized network analysis.

```
Command: nslookup networkwalks.com
```

5.6 dnsrecon

dnsrecon was used for DNS enumeration. The captured run completed its enumeration and reported an A-record observation. It also displayed messages concerning DNSSEC and SOA resolution/timeouts. These messages did not prevent the tool from completing the enumeration, and they illustrate that DNS reconnaissance tools may encounter resolver or server-side limitations while collecting information.

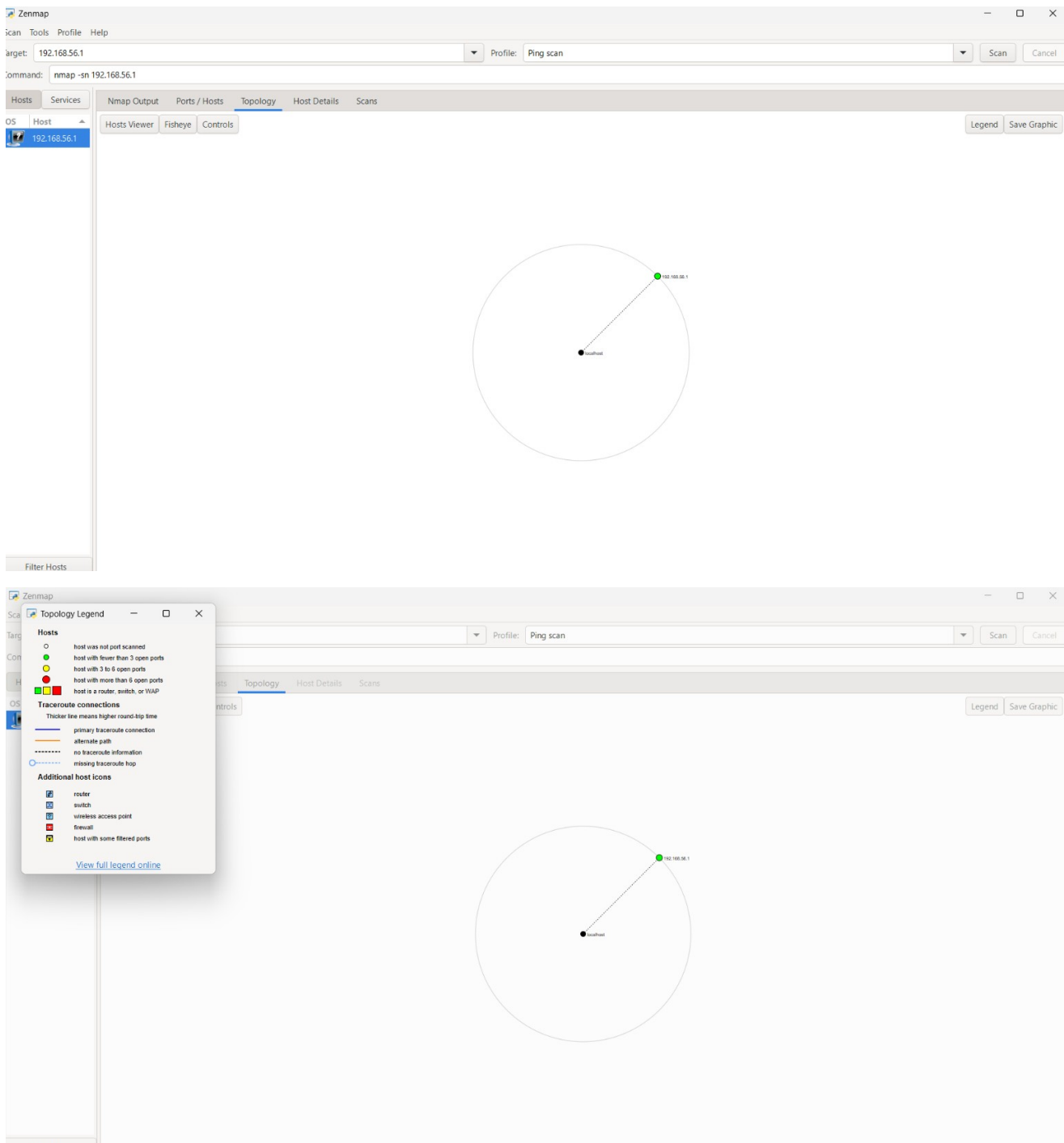
```
Command: dnsrecon -d networkwalks.com
```

6. W2-PM5 – Zenmap-Based Network Scanning

For the required essential module, Zenmap was used as a graphical interface for Nmap. The Ping Scan profile was selected and the target was set to the private lab host **192.168.56.1**.

The command shown by Zenmap was **nmap -sn 192.168.56.1**. The scan identified 192.168.56.1 as an active/reachable host and displayed it in the topology view. The -sn option performs host discovery without performing a normal port scan.

Zenmap configuration and topology evidence:



7. Findings and Learning Outcomes

The practical exercises demonstrated that reconnaissance is not a single command but a combination of complementary techniques. WHOIS and DNS tools provided domain and DNS context; WhatWeb and cURL provided web-server and HTTP information; WAFW00F provided an indication of possible web protection; and Zenmap demonstrated basic network host discovery.

Area	Learning outcome
Domain footprinting	Registration and domain-status information can be collected through WHOIS.
Web fingerprinting	Technology fingerprints can reveal server and application components.
HTTP analysis	Response headers and redirects provide useful web-server context.
DNS analysis	DNS resolution and enumeration help connect a domain name with network information.
Network discovery	Zenmap/Nmap can identify reachable hosts in an authorized lab network.
Security awareness	WAF/filtering behavior can affect reconnaissance results and must be interpreted carefully.

8. Conclusion

The Week 2 practical project was completed by combining W2-PM1 footprinting activities with W2-PM5 Zenmap-based network scanning. The exercises provided hands-on experience with reconnaissance tools commonly used to understand domains, web applications, DNS information, and reachable hosts. The results also demonstrated the importance of validating tool output and understanding limitations such as filtering, timeouts, and resolver behavior.

Overall, the project strengthened practical skills in information gathering, command-line tool usage, network discovery, result interpretation, and technical documentation.

9. Evidence and Submission Notes

The screenshots collected during the practical exercises should be retained as supporting evidence for the final submission. The Zenmap screenshots are included in this report. The W2-PM1 screenshots can be inserted in the corresponding sections if the submission requires every command to have a visual screenshot.

End of Report